Module 2

Load Flow Analysis

AC power flow analysis is basically a steady-state analysis of the AC transmission and distribution grid. Essentially, AC power flow method computes the steady state values of bus voltages and line power flows from the knowledge of electric loads and generations at different buses of the system under study. In this module, we will look into the power flow solution of the AC transmission grid only (the solution methodology of AC distribution grid will not be covered). Further, we will also study the power flow solution technique when an HVDC link is embedded into an AC transmission grid. Also, we will be considering only a balanced system in which the transmission lines and loads are balanced (the impedances are equal in all the three phases) and the generator produces balanced three phase voltages (magnitudes are equal in all the phases while the angular difference between any two phases is 120 degree).

2.1 Modeling of power system components

Basically, an AC transmission grid consists of, i) synchronous generator, ii) loads, iii) transformer and iv) transmission lines. For the purpose of power flow solution, synchronous generators are not represented explicitly, rather their presence in implicitly modeled. We will look into the implicit representation of synchronous generators a little later. However, the other three components are modeled explicitly and their representations are discussed below.

2.1.1 Loads

As we all know, loads can be classified into three categories; i) constant power, ii) constant impedance and iii) constant current. However, within the normal operating range of the voltage almost all the loads behave as constant power loads. As the objective of the AC power flow analysis is to compute the normal steady-state values of the bus voltages, the loads are always represented as constant power loads. Hence, at any bus 'k' (say), the real and reactive power loads are specified as 100 MW and 50 MVAR (say) respectively. An important point needs to be mentioned here. As the loads are always varying with time (the customers are always switching 'ON' and 'OFF' the loads), any specific value of load (MW and/or MVAR) is valid only at a particular time instant. Hence, AC

power flow analysis is always carried out for the load and generator values at a particular instant.

2.1.2 Transmission line

In a transmission grid, the transmission lines are generally of medium length or of long length. A line of medium length is always represented by the nominal-π model as shown in Fig. 2.1, where $\bar{z}$ is the total series impedance of the line and B_c is the total shunt charging susceptance of the line. On the other hand, a long transmission line is most accurately represented by its distributed parameter model. However, for steady-state analysis, a long line can be accurately represented by the equivalent-π model, which predicts accurate behavior of the line with respect to its terminal measurements taken at its two ends. The equivalent-π model is shown in Fig. 2.2.

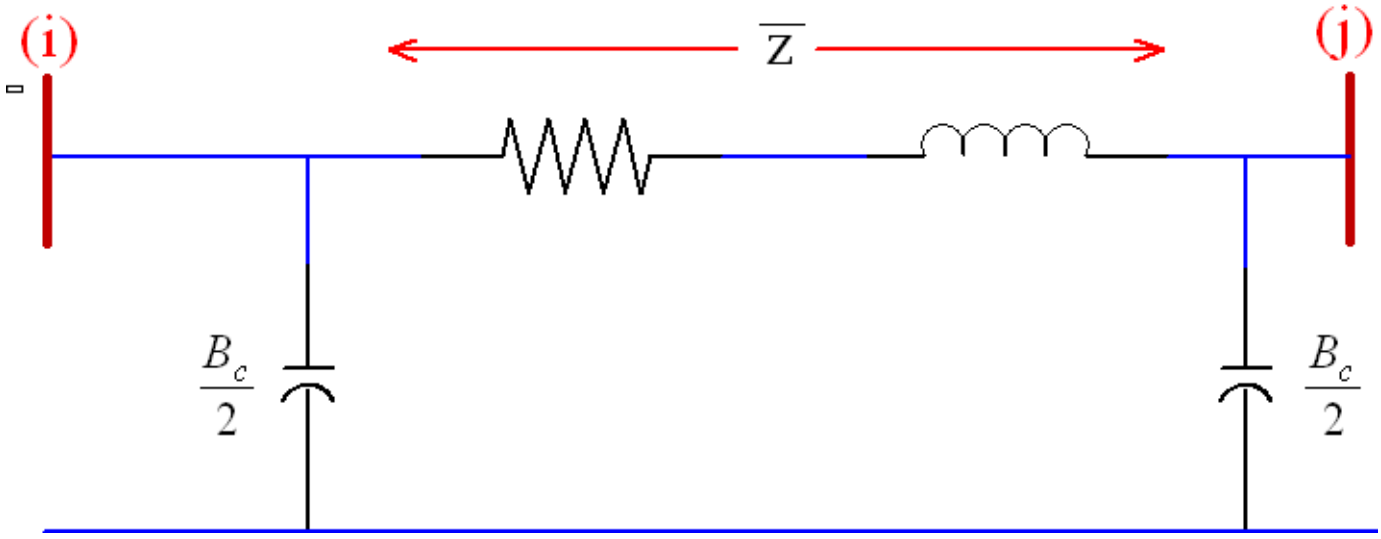


Figure 2.1: Normal π model of a line connected between buses 'i' and 'j'

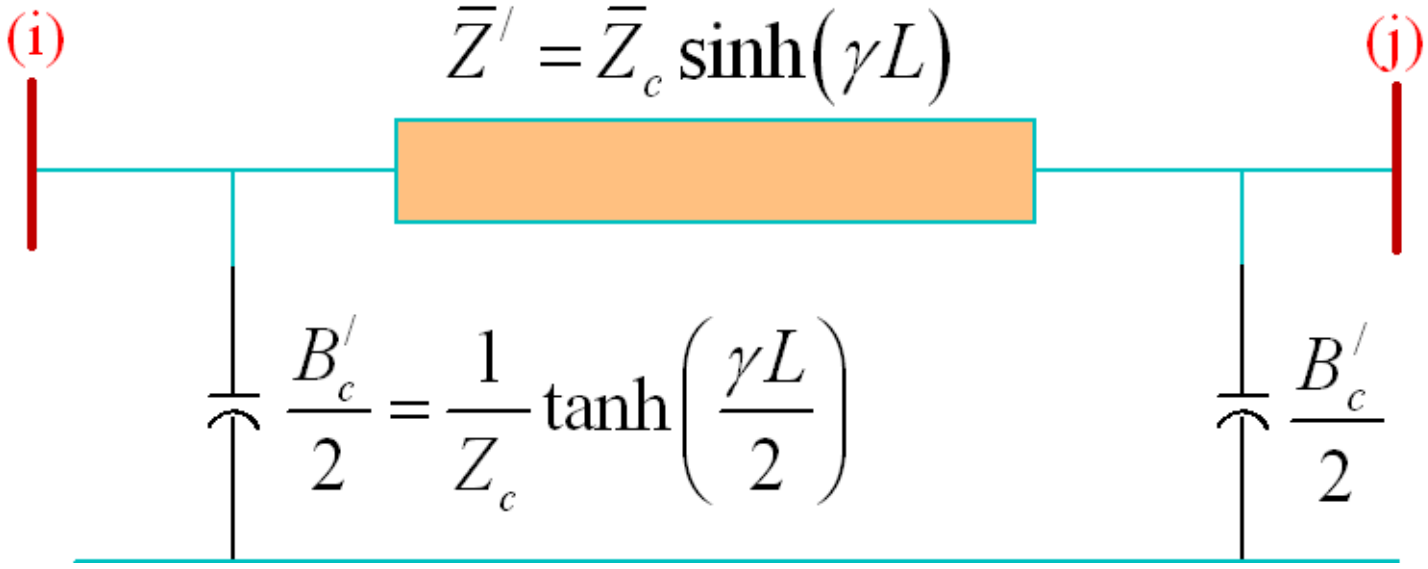


Figure 2.2: Equivalent π model of a long transmission line connected between buses 'i' and 'j'

In Fig. 2.2,

$\bar{z}_c = \dfrac{\bar{z}}{\bar{y}}$ is the characteristic impedance of the line
$\gamma = \sqrt{\bar{z}\bar{y}}$ is the propagation constant
$\bar{z}$ = series impedance of the line per unit length
$\bar{y}$ = shunt admittance of the line per unit length
L = length of the line

Hence, for power system analysis, a transmission line (medium or long) is always represented by a π circuit.

2.1.3 Transformer

For power system steady-state and fault studies, generally the exciting current of the transformer is neglected as it is quite low compared to the normal load current flowing through the transformer. Therefore, a two winding transformer connected between buses 'i' and 'j' is represented by its per unit leakage impedance as shown in Fig. 2.3.

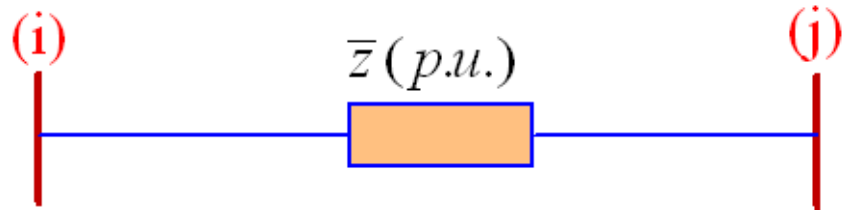


Figure 2.3: Equivalent Equivalent circuit of a two winding transformer

It is to be noted that in Fig. 2.3, the transformer tap ratio is 1:1. For a regulating transformer with transformation ratio 1:t, the equivalent circuit of the transformer is shown in Fig. 2.4.

Sometimes the transformer ratio is also represented as a:1. In that case, the equivalent circuit is as shown in Fig. 2.5.

Please note that in Figs. 2.4 and 2.5, the quantities 't' and 'a' are real (i.e. the transformer is changing only the voltage magnitude, not its angle). Further, in these two figures, the quantity $\bar{y}$ is the per unit admittance of the transformer. Also, Fig. 2.5 can be derived from Fig. 2.4 by noting $t = 1/a$ and by interchanging the buses 'i' and 'j'.

With the models of above components in place, we are now in a position to start systematic study of an 'n' bus power system. Towards that goal, we first must understand the concept of injected power and injected current, which is our next topic.

2.2 Concept of injected power and current

As the name suggests, the injected power (current) indicates the power (current) which is fed 'in' to a bus. To understand this concept, let us consider Fig. 2.6. In part (a) of this figure, a generator is connected at bus 'k' supplying both real and reactive power to the bus and thus, the injected real and reactive power are taken to be equal to the real (reactive) power supplied by the generator. The

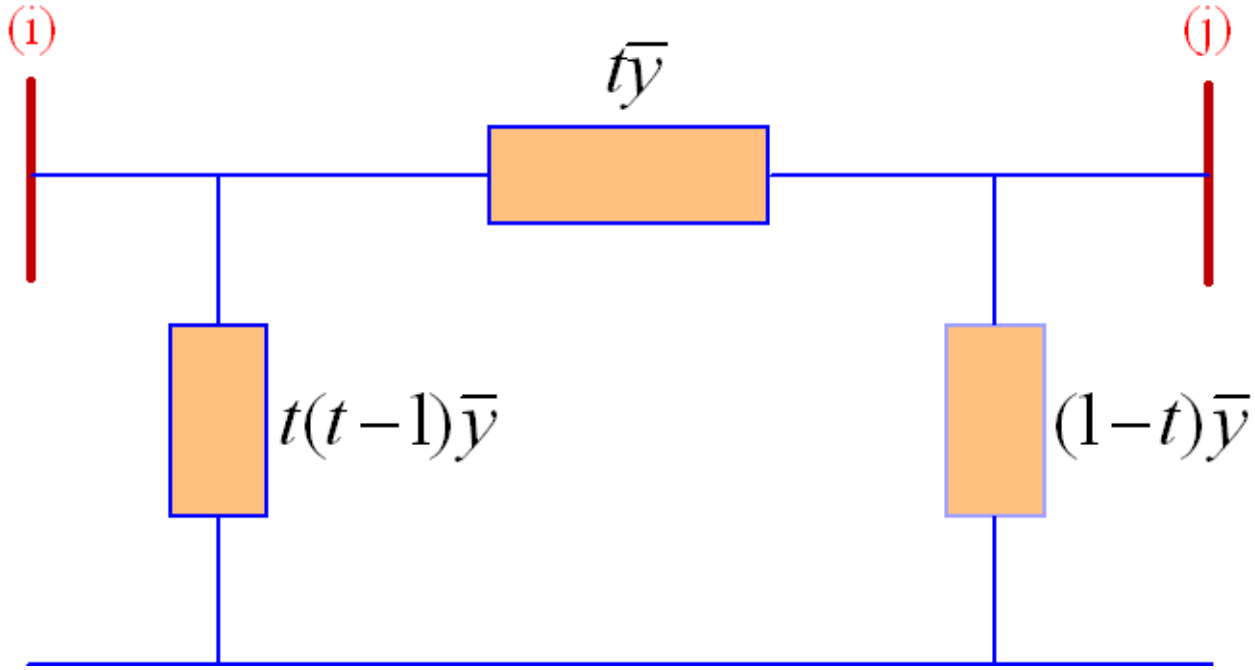


Figure 2.4: Equivalent circuit of a regulating transformer with transformation ratio 1:t

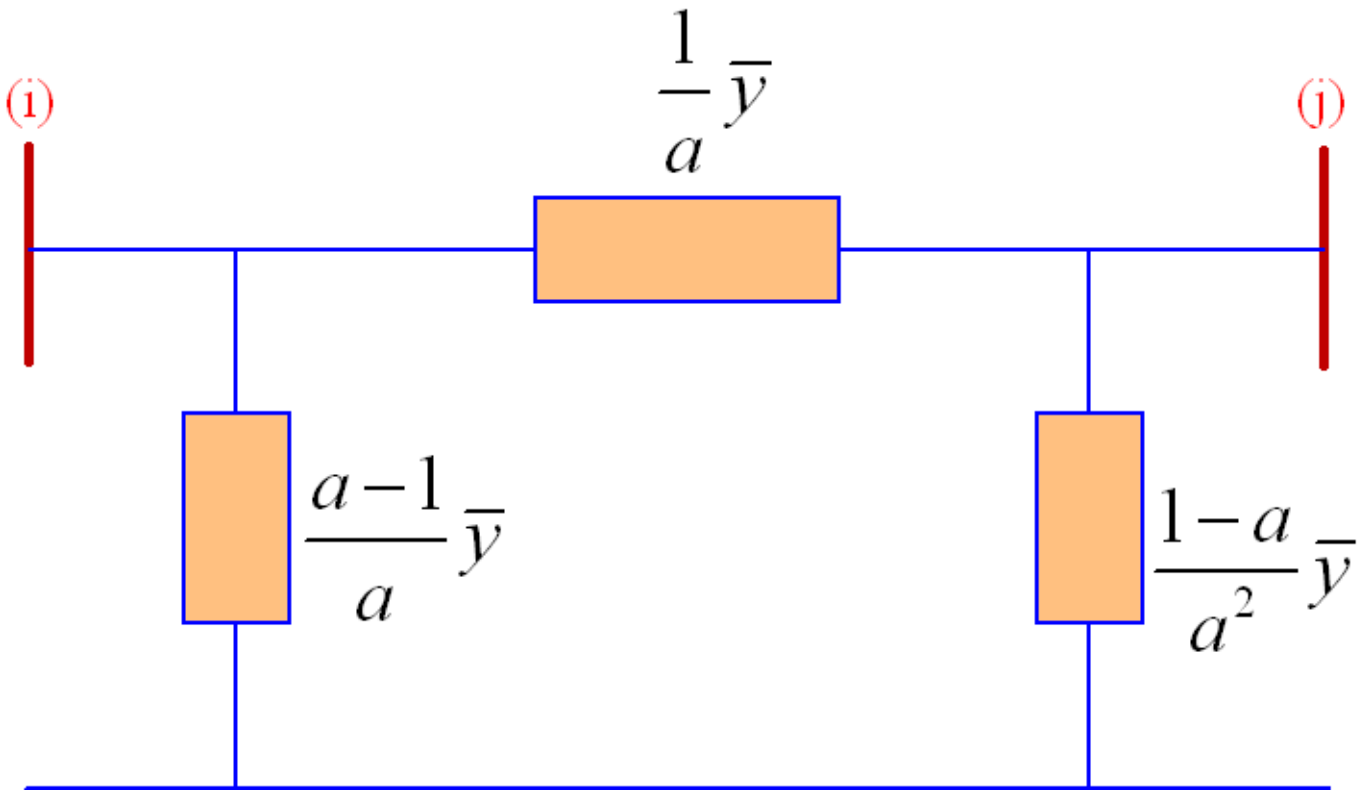


Figure 2.5: Equivalent circuit of a regulating transformer with transformation ratio a:1

corresponding injected current is also taken to be equal to the current supplied by the generator. On the other hand, for a load connected to bus 'k' (as shown in Fig. 2.6(b)), physically the real (reactive) power consumed by the load flows away from the bus and thus, the injected real (reactive) power is taken to be the negative of the real (reactive) power consumed by the load. Similarly, the corresponding injected current $\bar{I}_k$ is also taken as the negative of the load current. If both a generator

and a load are connected at a particular bus (as depicted in Fig. 2.6(c)), then the net injected real (reactive) power supplied to the bus is equal to the generator real (reactive) power minus the real (reactive) power consumed by the load. Similarly, the net injected current in this case is taken to be the difference of the generator current and the load current.

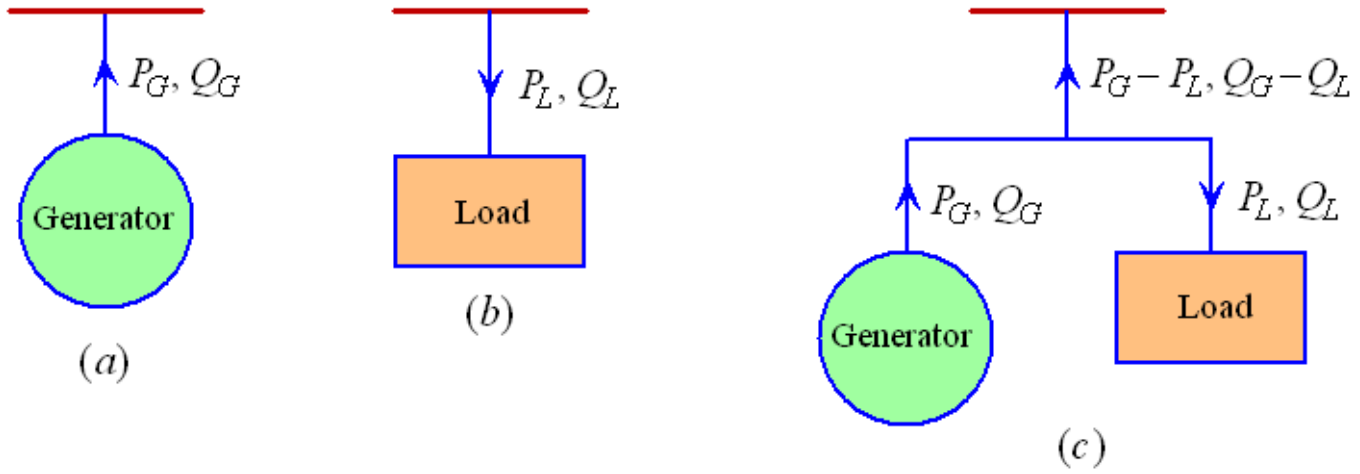


Figure 2.6: Illustration of injected power

To summarize, if P_k, Q_k, and $\bar{I}_k$ denote the injected real power, reactive power and complex current at bus 'k' respectively,

- $P_k = P_G$; $Q_k = Q_G$ and $\bar{I}_k = \bar{I}_G$ if only a generator is connected to the bus 'k'.
- $P_k = -P_L$; $Q_k = -Q_L$ and $\bar{I}_k = -\bar{I}_L$ if only a load is connected to the bus 'k'.
- $P_k = P_G - P_L$; $Q_k = Q_G - Q_L$ and $\bar{I}_k = \bar{I}_G - \bar{I}_L$ if both generator and load are connected to the bus 'k'.
- $P_k = 0$; $Q_k = 0$; $\bar{I}_k = 0$ If neither generator nor load is connected to the bus 'k'.

With this concept of injected power and current, we are now in a position to start analysis of any general 'n' bus power system. The first step towards this goal is to derive the bus admittance matrix, which we will take up next.

2.3 Formation of bus admittance matrix ($\bar{\mathbf{Y}}_{\mathbf{BUS}}$)

Let us consider a 5 bus network as shown in Fig. 2.7. In this network, all the transmissions are represented by π models. Therefore, the equivalent circuit of the above network is shown in Fig. 2.8.

In Fig. 2.8, $\bar{I}_k$; k = 1, 2, 3, 4, 5 are the injected currents at bus 'k'. Further, the quantity $\bar{y}_{ij}$ denotes the series admittance of the line 'i-j' whereas the quantity $\bar{y}_{ijs}$ denotes the half line charging susceptance of the line 'i-j'. Now applying 'KCL' at each bus 'k' one obtains,

$$\bar{I}_1 = \bar{y}_{T1}(\bar{V}_1 - \bar{V}_2) = \bar{y}_{T1}\bar{V}_1 - \bar{y}_{T1}\bar{V}_2 \tag{2.1}$$

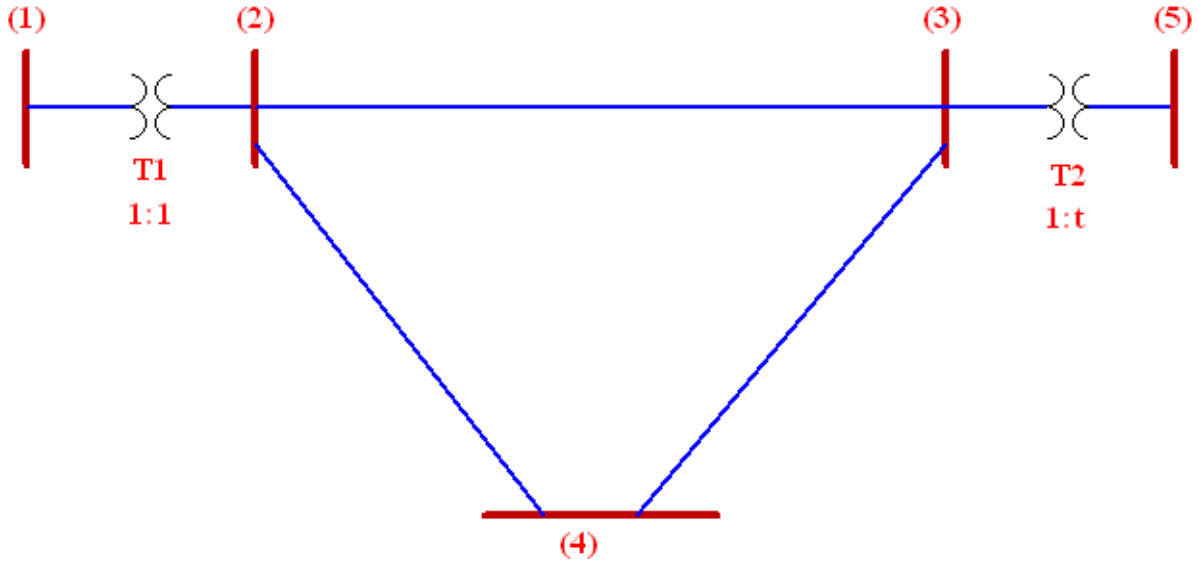


Figure 2.7: A sample 5 bus network

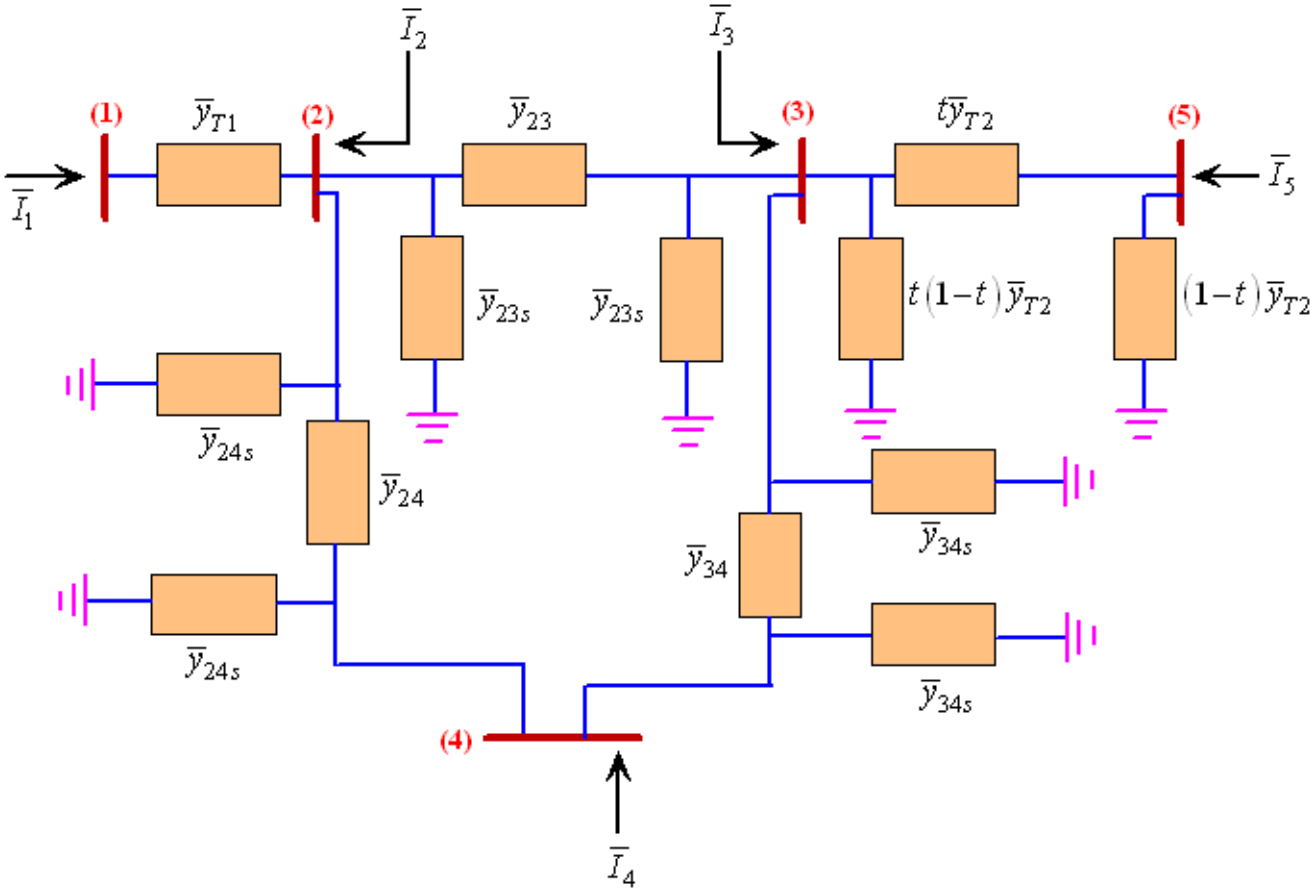


Figure 2.8: Equivalent circuit of Fig. 2.7

$$\begin{aligned}\bar{I}_2 &= \bar{y}_{T1}(\bar{V}_2 - \bar{V}_1) + \bar{V}_2\bar{y}_{23s} + (\bar{V}_2 - \bar{V}_3)\bar{y}_{23} + \bar{V}_2\bar{y}_{24s} + (\bar{V}_2 - \bar{V}_4)\bar{y}_{24} \\ &= -\bar{y}_{T1}\bar{V}_1 + (\bar{y}_{T1} + \bar{y}_{23s} + \bar{y}_{23} + \bar{y}_{24s} + \bar{y}_{24})\bar{V}_2 - \bar{y}_{23}\bar{V}_3 - \bar{y}_{24}\bar{V}_4 \qquad (2.2)\end{aligned}$$

$$\begin{aligned}\bar{I}_3 &= (\bar{V}_3-\bar{V}_2)\bar{y}_{23}+\bar{V}_3\bar{y}_{23s}+(\bar{V}_3-\bar{V}_5)t\bar{y}_{T2}+t(t-1)\bar{y}_{T2}\bar{V}_3+(\bar{V}_3-\bar{V}_4)\bar{y}_{34}+\bar{V}_3\bar{y}_{34s}\\ &= -\bar{V}_2\bar{y}_{23}+\left\{\bar{y}_{23}+\bar{y}_{23s}+t\bar{y}_{T2}+t(t-1)\bar{y}_{T2}+\bar{y}_{34}+\bar{y}_{34s}\right\}\bar{V}_3-\bar{y}_{34}\bar{V}_4-t\bar{y}_{T2}\bar{V}_5\end{aligned} \tag{2.3}$$

$$\begin{aligned}\bar{I}_4 &= (\bar{V}_4-\bar{V}_2)\bar{y}_{24}+\bar{y}_{24s}\bar{V}_4+(\bar{V}_4-\bar{V}_3)\bar{y}_{34}+\bar{y}_{34s}\bar{V}_4\\ &= -\bar{V}_2\bar{y}_{24}-\bar{V}_3\bar{y}_{34}+\left(\bar{y}_{24}+\bar{y}_{24s}+\bar{y}_{34}+\bar{y}_{34s}\right)\bar{V}_4\end{aligned} \tag{2.4}$$

$$\begin{aligned}\bar{I}_5 &= (\bar{V}_5-\bar{V}_3)t\bar{y}_{T2}+(1-t)\bar{y}_{T2}\bar{V}_5\\ &= -\bar{V}_3t\bar{y}_{T2}+\left\{t\bar{y}_{T2}+(1-t)\bar{y}_{T2}\right\}\bar{V}_5\end{aligned} \tag{2.5}$$

Equations (2.1) - (2.5) can be represented in a matrix form as,

$$\begin{bmatrix}\bar{I}_1\\ \bar{I}_2\\ \bar{I}_3\\ \bar{I}_4\\ \bar{I}_5\end{bmatrix}=\begin{bmatrix}\bar{Y}_{11} & \bar{Y}_{12} & \bar{Y}_{13} & \bar{Y}_{14} & \bar{Y}_{15}\\ \bar{Y}_{21} & \bar{Y}_{22} & \bar{Y}_{23} & \bar{Y}_{24} & \bar{Y}_{25}\\ \bar{Y}_{31} & \bar{Y}_{32} & \bar{Y}_{33} & \bar{Y}_{34} & \bar{Y}_{35}\\ \bar{Y}_{41} & \bar{Y}_{42} & \bar{Y}_{43} & \bar{Y}_{44} & \bar{Y}_{45}\\ \bar{Y}_{51} & \bar{Y}_{52} & \bar{Y}_{53} & \bar{Y}_{54} & \bar{Y}_{55}\end{bmatrix}\begin{bmatrix}\bar{V}_1\\ \bar{V}_2\\ \bar{V}_3\\ \bar{V}_4\\ \bar{V}_5\end{bmatrix} \tag{2.6}$$

Where, $\bar{Y}_{11}=\bar{y}_{T1}$; $\bar{Y}_{12}=-\bar{y}_{T1}$; $\bar{Y}_{13}=\bar{Y}_{14}=\bar{Y}_{15}=0$; $\bar{Y}_{21}=-\bar{y}_{T1}$;
$\bar{Y}_{22}=\left(\bar{y}_{T1}+\bar{y}_{23s}+\bar{y}_{23}+\bar{y}_{24s}+\bar{y}_{24}\right)$; $\bar{Y}_{23}=-\bar{y}_{23}$; $\bar{Y}_{24}=-\bar{y}_{24}$; $\bar{Y}_{25}=0$;
$\bar{Y}_{31}=0$; $\bar{Y}_{32}=-\bar{y}_{23}$; $\bar{Y}_{33}=\left\{\bar{y}_{23}+\bar{y}_{23s}+t\bar{y}_{T2}+t(t-1)\bar{y}_{T2}+\bar{y}_{34}+\bar{y}_{34s}\right\}$;
$\bar{Y}_{34}=-\bar{y}_{34}$; $\bar{Y}_{35}=-t\bar{y}_{T2}$; $\bar{Y}_{41}=0$; $\bar{Y}_{42}=-\bar{y}_{24}$; $\bar{Y}_{43}=-\bar{y}_{34}$;
$\bar{Y}_{44}=\left(\bar{y}_{24}+\bar{y}_{24s}+\bar{y}_{34}+\bar{y}_{34s}\right)$; $\bar{Y}_{45}=0$; $\bar{Y}_{51}=\bar{Y}_{52}=0$;
$\bar{Y}_{53}=-t\bar{y}_{T2}$; $\bar{Y}_{54}=0$; $\bar{Y}_{55}=\left\{t\bar{y}_{T2}+(1-t)\bar{y}_{T2}\right\}$

Equation (2.6) can be written as,

$$\boxed{\bar{\mathbf{I}}_{\mathbf{BUS}}=\bar{\mathbf{Y}}_{\mathbf{BUS}}\bar{\mathbf{V}}_{\mathbf{BUS}}} \tag{2.7}$$

Where,

$\bar{\mathbf{I}}_{\mathbf{BUS}}=\left[\bar{I}_1,\ \bar{I}_2\cdots\bar{I}_5\right]^T\rightarrow(5\times1)$ is the vector of bus injection currents

$\bar{\mathbf{V}}_{\mathbf{BUS}}=\left[\bar{V}_1,\ \bar{V}_2\cdots\bar{V}_5\right]^T\rightarrow(5\times1)$ is the vector of bus voltages measured with respect to the ground

$\bar{\mathbf{Y}}_{\mathbf{BUS}}\rightarrow(5\times5)$ is the bus admittance matrix

Furthermore, from the elements of the $\bar{\mathbf{Y}}_{\mathbf{BUS}}$ it can be observed that for $i=1,2,\cdots\cdots5$;

$\bar{Y}_{ii}$ = sum total of all the admittances connected at bus 'i'

$\bar{Y}_{ij}$ = negative of the admittance connected between bus 'i' and 'j' (if these two buses are physically connected with each other)

$\bar{Y}_{ij}=0$; if there is no physical connection between buses 'i' and 'j'

Similarly, for a 'n' bus power system, the relation given in equation (2.7) holds good, where,

$\bar{\mathbf{I}}_{\mathbf{BUS}} = \left[\bar{I}_1,\ \bar{I}_2 \cdots \bar{I}_n\right]^T \rightarrow (n \times 1)$ is the vector of bus injection currents

$\bar{\mathbf{V}}_{\mathbf{BUS}} = \left[\bar{V}_1,\ \bar{V}_2 \cdots \bar{V}_n\right]^T \rightarrow (n \times 1)$ is the vector of bus voltages

$\bar{\mathbf{Y}}_{\mathbf{BUS}} \rightarrow (n \times n)$ is the bus admittance matrix

Furthermore, the elements of the $\bar{\mathbf{Y}}_{\mathbf{BUS}}$ matrix are calculated in the same way as described above.

So far, we have considered the formation of the $\bar{\mathbf{Y}}_{\mathbf{BUS}}$ matrix when there is no mutual coupling among the elements of the network. In the next lecture, we will look into the procedure for forming the $\bar{\mathbf{Y}}_{\mathbf{BUS}}$ matrix in the presence of mutual coupling between the elements.